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Avadi, Chennai

Vel Tech Rangarajan Dr. Sagunthala R&D Institute of Science and Technology

School of Computing

Department of CSE-Cyber Security

10211CC224 / FULL STACK APPLICATION DEVELOPMENT (JAVA)

PROJECT REVIEW : MILESTONE PROJECT – 1

Date: 02-04-2026

Smart Study Planner

SDG: SDG No – SDG Title(Example: SDG 4 – Quality Education)

Presented By:

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Courses Handling Faculty:

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INTRODUCTION

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- **Background:**

The shift to digital education leaves students overwhelmed with scattered resources (PDFs, PPTs, web links) across multiple subjects.

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- **The Problem:**

Manual file tracking across different platforms causes academic disorganization, reduced productivity, and missed deadlines.

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- **The Need:**

Students require specialized tools—not just general cloud storage—to actively streamline exam preparation and daily tasks.

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- **Our Solution:**

The Smart Study Planner—a full-stack Java application centralizing secure document storage, subject organization, and interactive revision tools in one structured platform.

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PROBLEM STATEMENT

- **Fragmented Tools:** Reliance on generic cloud storage or manual folders causes significant delays in finding specific documents during exam prep.
- **Lack of Context:** Existing tools fail to link study materials directly to subject-wise schedules or tasks.
- **Missing Integration:** There is no single platform that combines simple document storage with active study aids (like revision cards and reminders).
- **The Impact:** Students suffer from disorganized study environments, leading to decreased productivity and missed deadlines.
- **The Clear Need:** A simple, secure, and automated system designed specifically to streamline resources and maximize academic focus.

OBJECTIVE(S)

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- **Centralized Resource Management:** Design and develop a full-stack Spring Boot application that organizes and centralizes study materials through a subject-linked document system.
- **Interactive Study Tools:** Implement a dynamic Study Board featuring visual revision cards and automated task reminders to boost academic productivity and streamline exam prep.

PROPOSED SYSTEM

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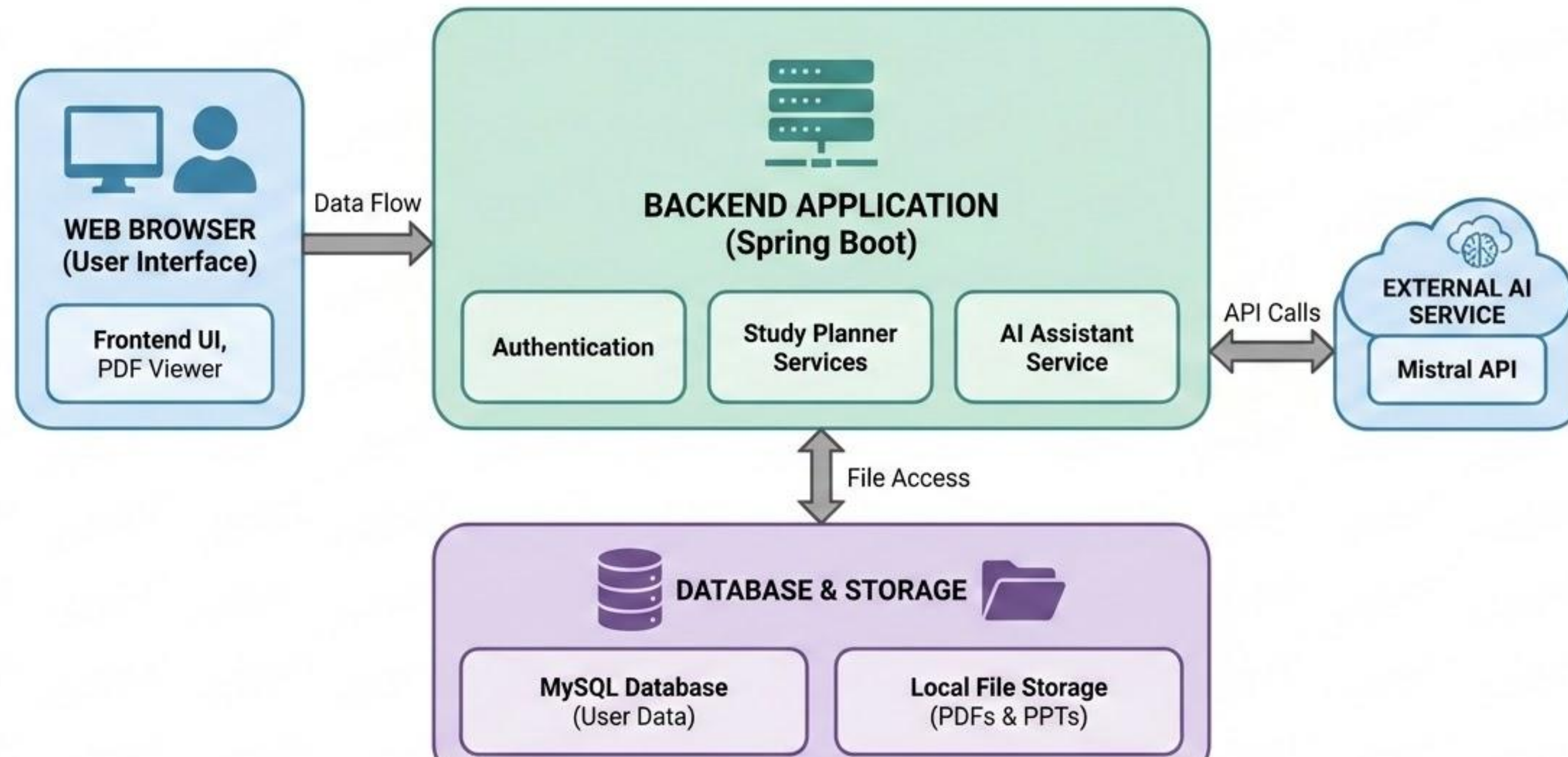
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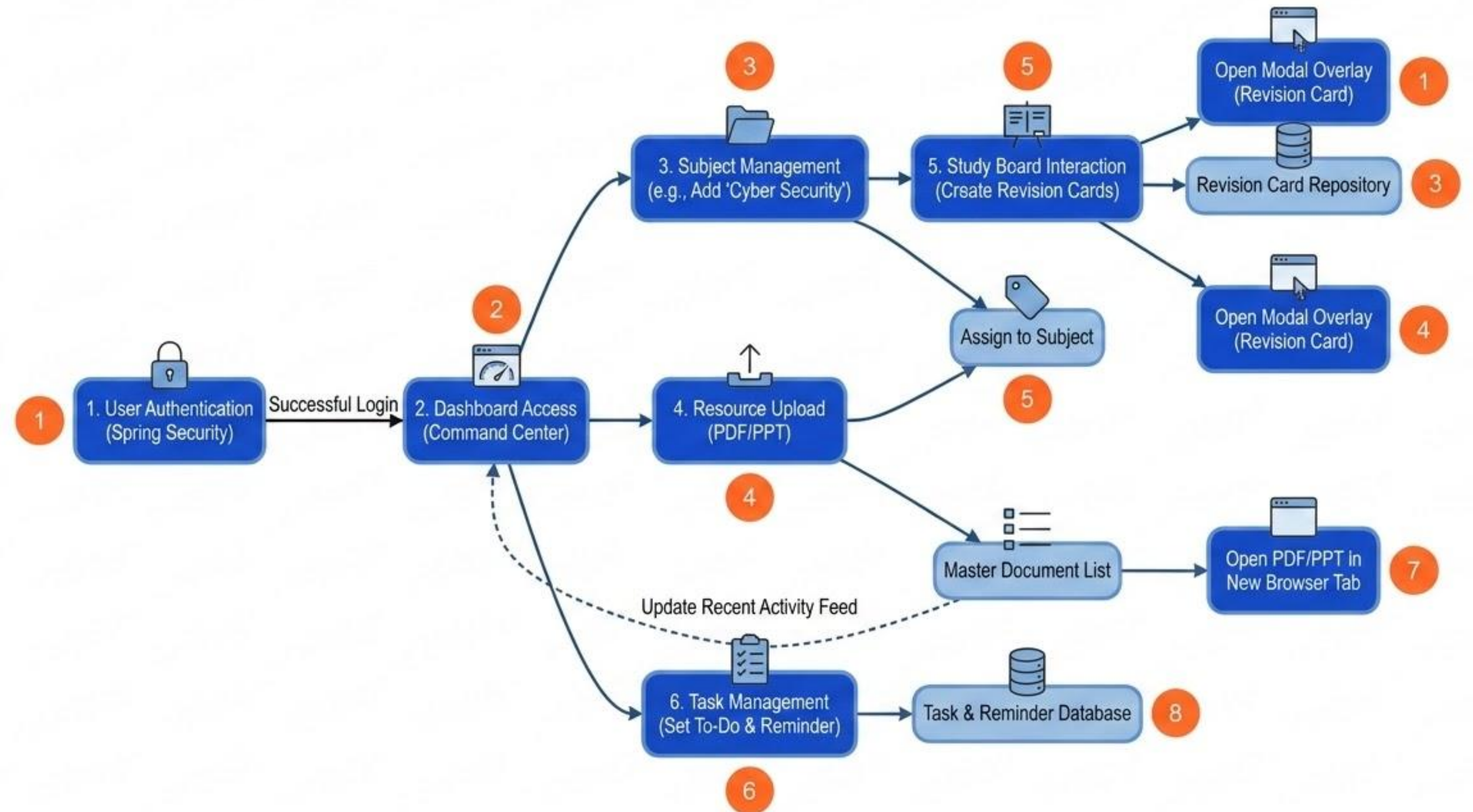
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- **Comprehensive Hub:** Centralizes subjects, documents, and tasks using Spring Boot & MySQL.
- **Intelligent Categorization:** Automatically links uploaded PDFs & PPTs to specific subjects for instant retrieval.
- **Interactive Study Boards:** Features visual "Revision Cards" (supporting text & images) for active exam prep.
- **Integrated Productivity:** Built-in to-do lists and automated reminders to effectively manage deadlines.
- **Seamless Experience:** In-browser document viewing and modal overlays for quick editing without leaving the dashboard.
- **Live Activity Tracking:** Real-time action feed providing a smooth "pick-up-where-you-left-off" experience.
- **Robust Security:** Spring Security authentication and protected profile management for strict data privacy.

PROPOSED SYSTEM ARCHITECTURE



WORKFLOW



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TECHNOLOGIES USED

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- **Frontend:**

- HTML, CSS, Postman

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- **Backend:**

- Spring Boot, Java (JDK 17+), Spring Web (REST API).

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- **Database:**

- MySQL

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- Tools: Postman, IntelliJ IDEA / Eclipse, Maven (build tool).

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- **Hardware:**

- Minimum 4GB RAM,

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- Laptop/Desktop System.

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- **Software Requirements:**

- Java JDK 17+,

- Spring Boot,

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- MySQL,

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- IntelliJ / Eclipse IDE,

- Postman

MODULE DESCRIPTION

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Module 1: User Authentication & Security

- Manages secure user registration and login using Spring security
- Provides profile management, including data updates and password encryption

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Module 2: Content & Resource Management

- Handles the core logic for adding, editing, and deleting Subjects
- Facilitates the Smart Upload of PDFs and PPTs, linking files directly to their respective subject categories

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Module 3: Interactive Revision Board :

- Implements the Revision Card system, allowing users to create study aids with text and images.
- Includes the Modal Overlay feature for quick content viewing and live editing without page reloads.

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Module 4: Productivity & Activity Tracker

- Manages the To-Do List and automated reminders for academic tasks.
- Generates the Recent Activity Feed on the Dashboard to provide a chronological log of user progress

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Module 5: AI powers assist

- Integrates the mistral api to provide intelligent aware study assistance

IMPLEMENTATION SCREENSHOTS

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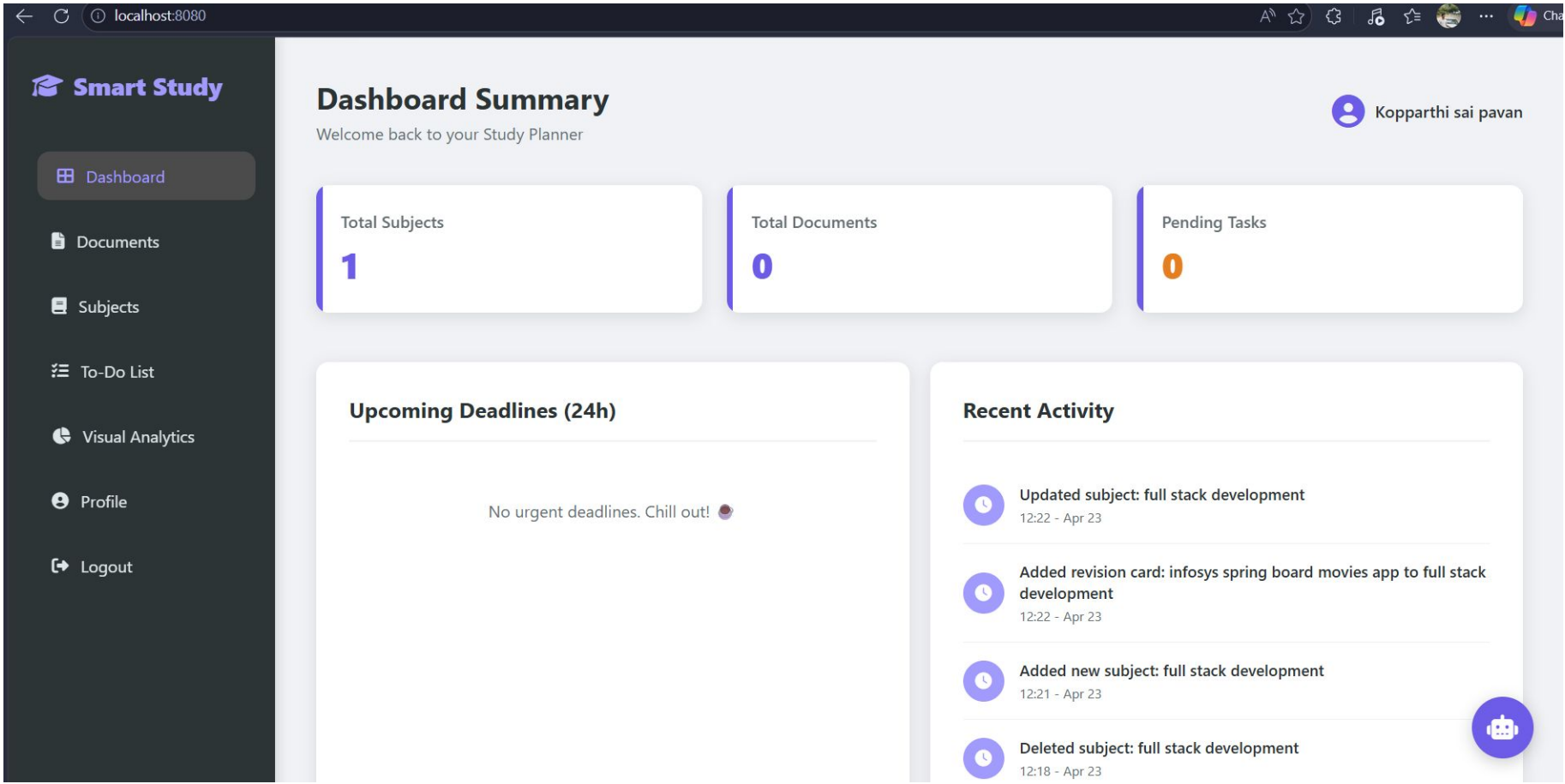
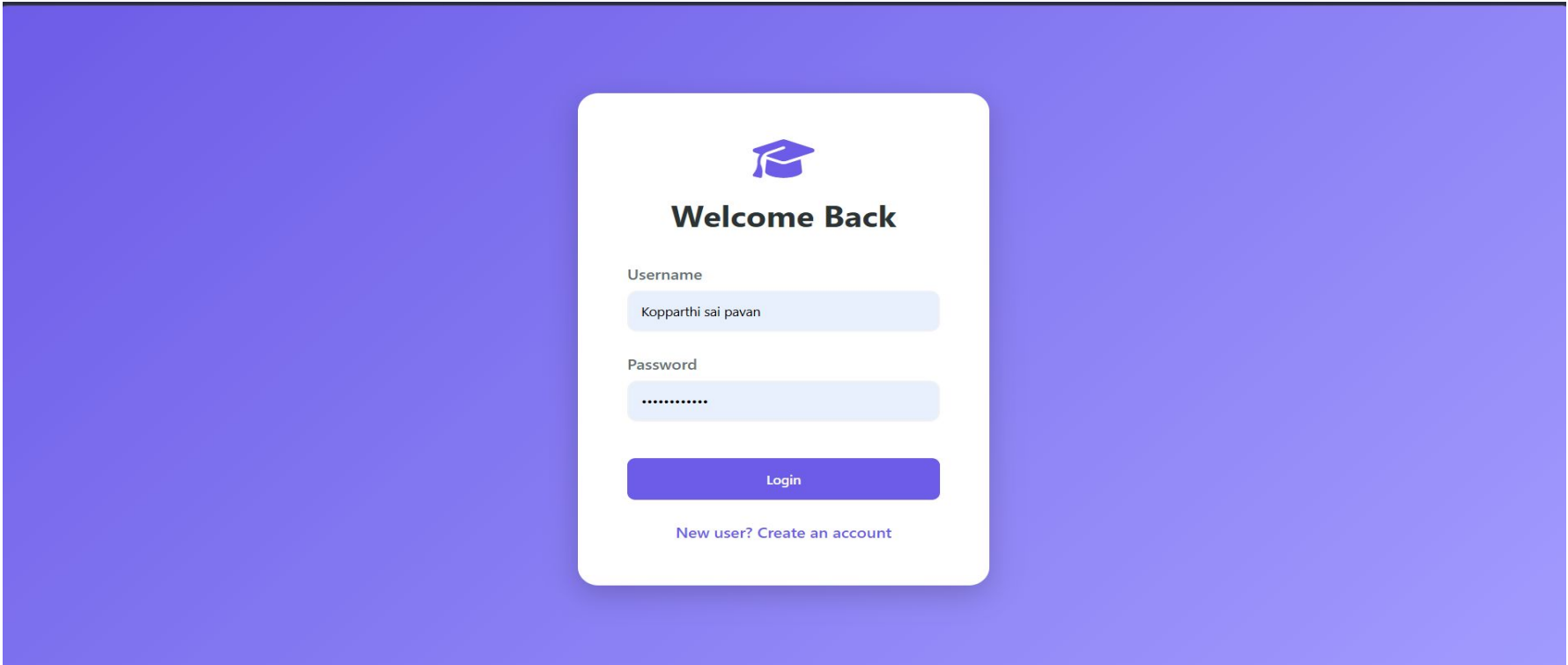
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IMPLEMENTATION SCREENSHOTS

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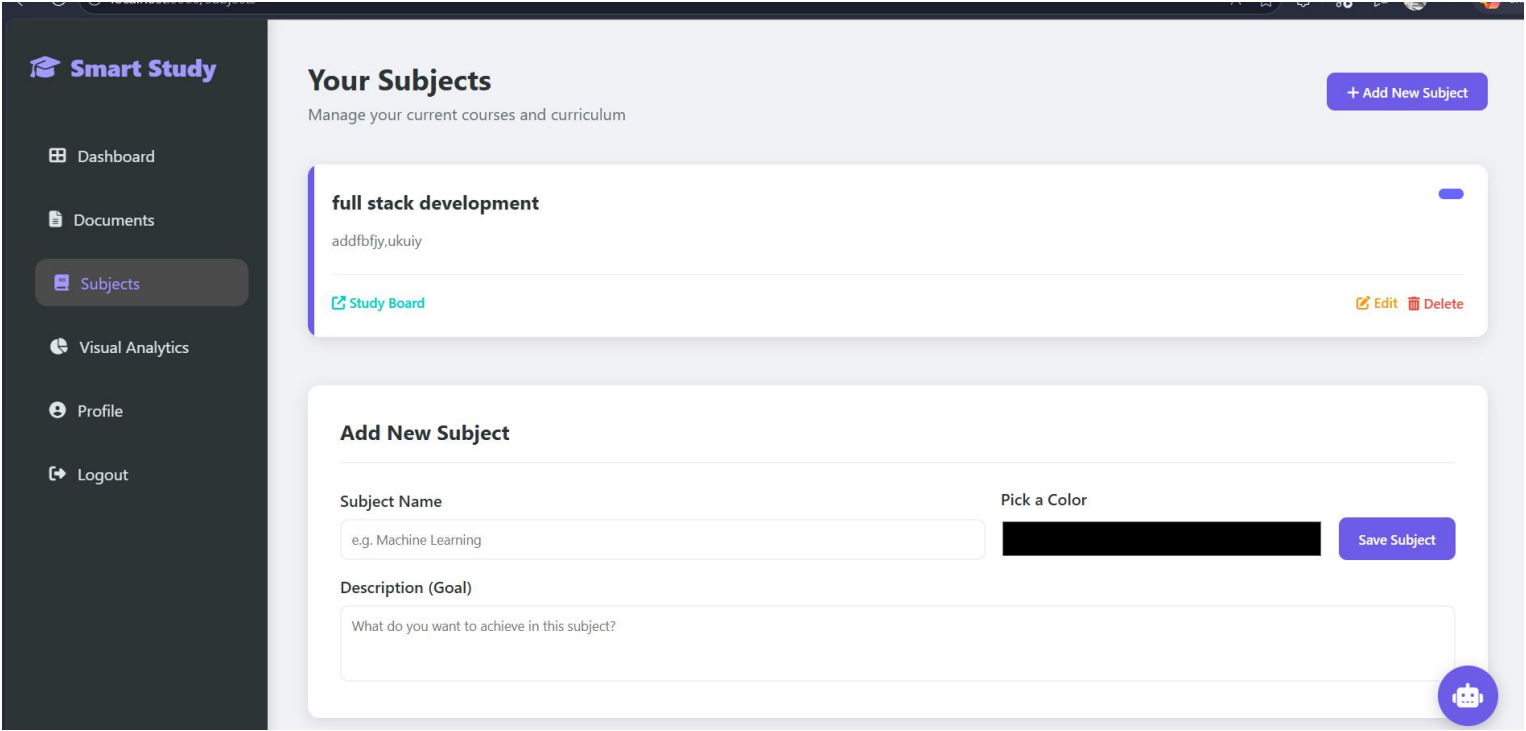
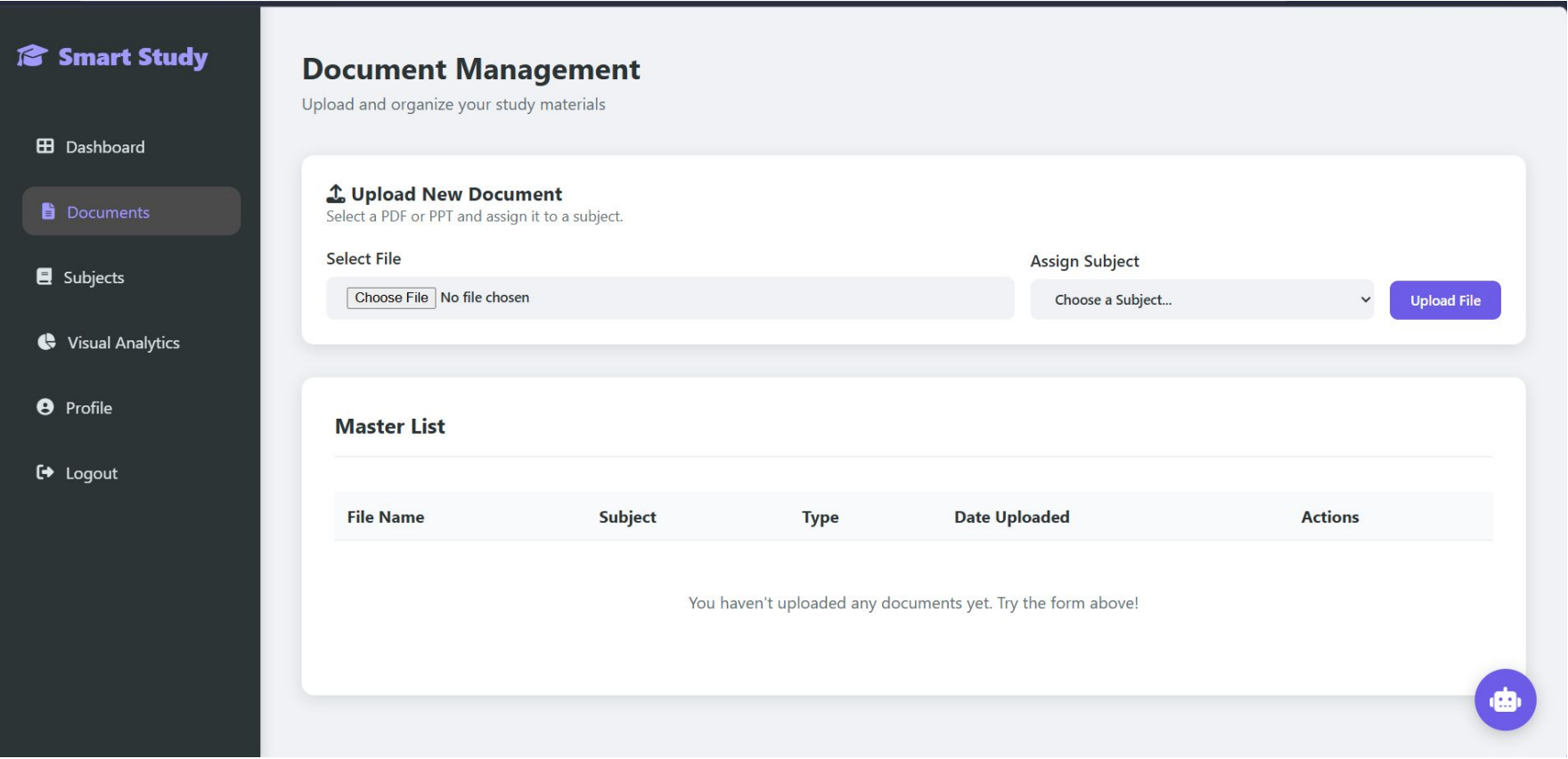
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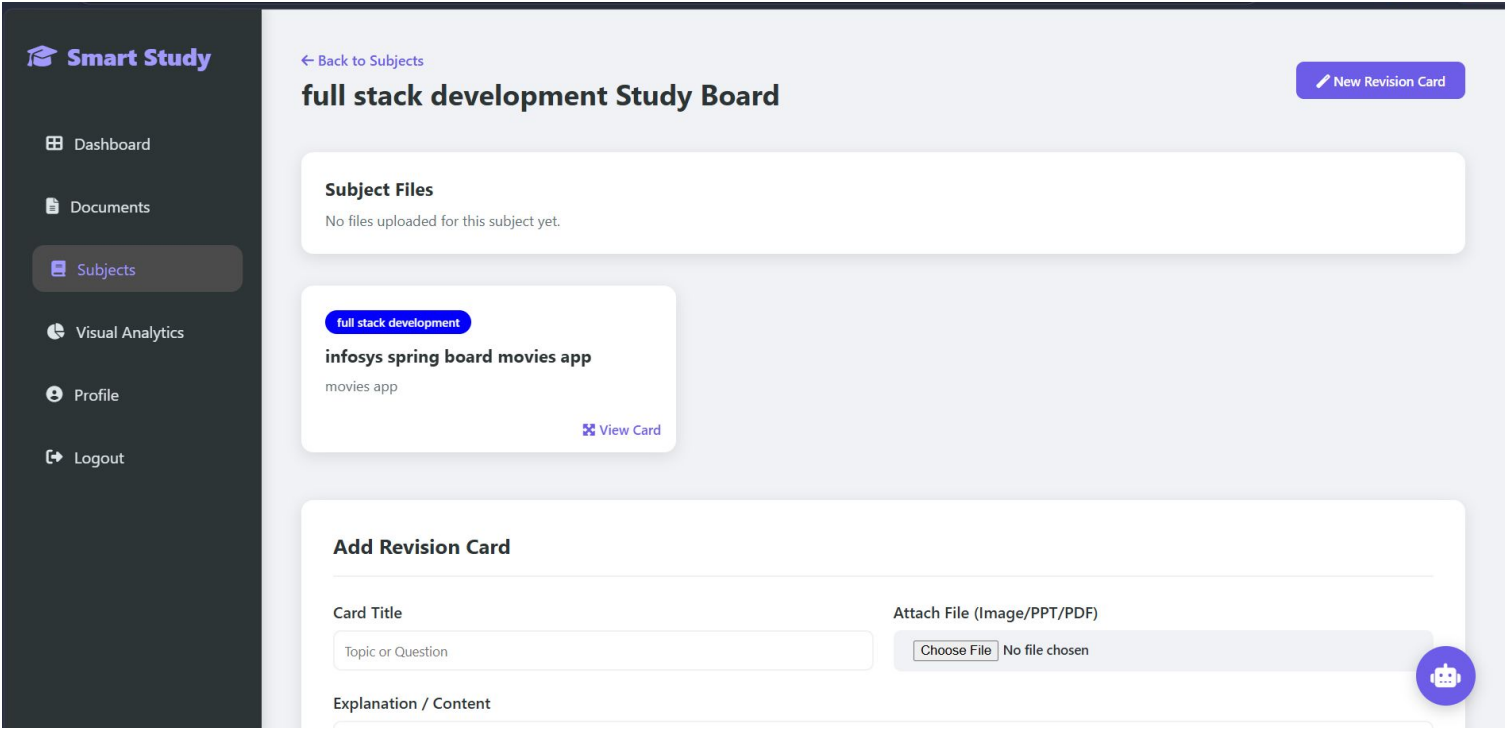
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IMPLEMENTATION SCREENSHOTS

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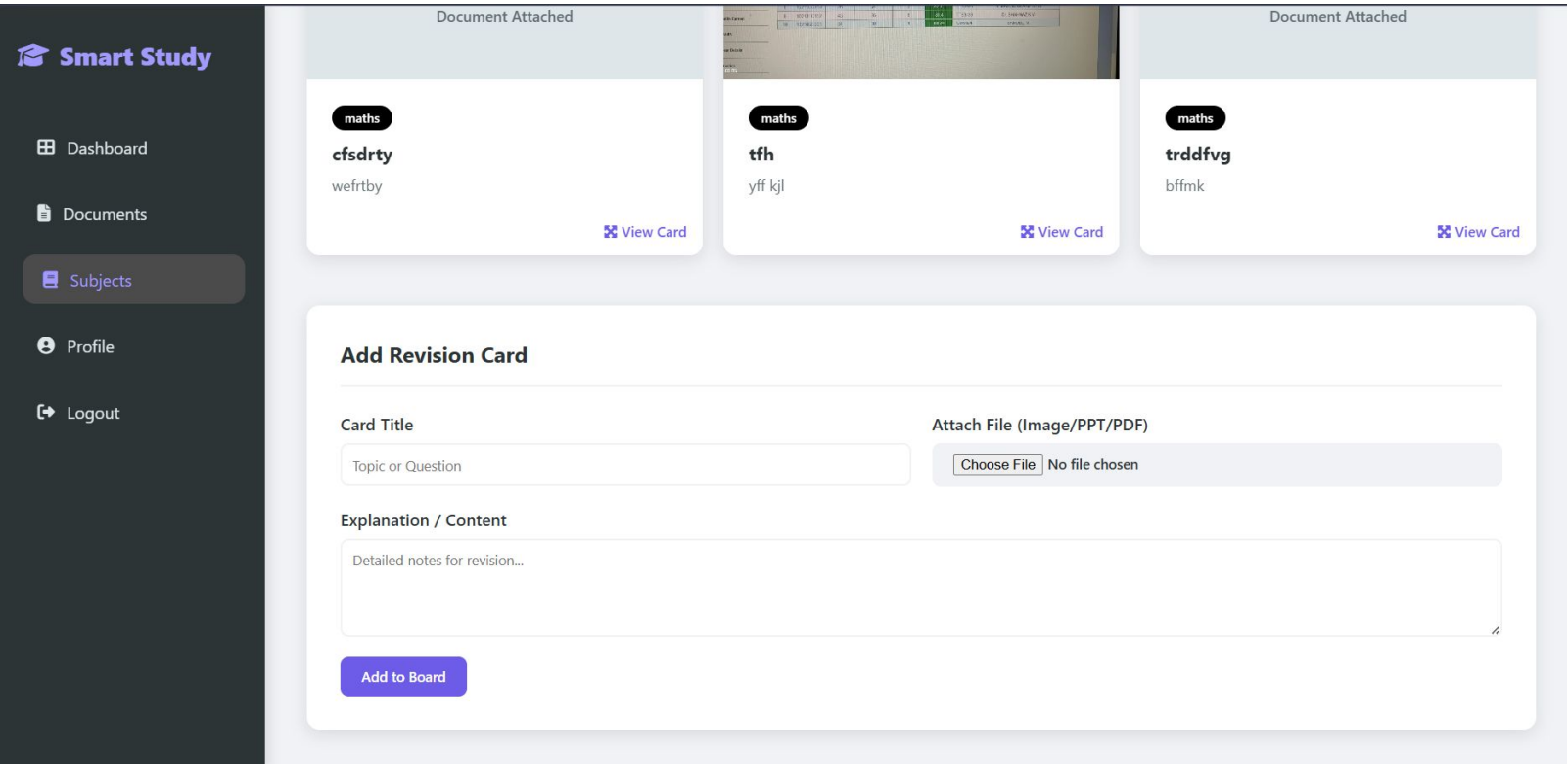
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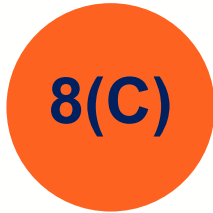
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ADVANTAGES & LIMITATIONS

ADVANTAGES:

- **Centralized Learning Hub:** Seamlessly integrates document storage, subject tracking, and task management into one structured platform.
- **AI-Powered Assistance (NEW):** Integrates the Mistral AI API to provide a stateful, intelligent study tutor that helps answer questions and summarize concepts in real-time.
- **Smart Study Tools:** Revision Card Modals and direct document mapping provide a much more focused experience than generic file explorers.
- **Professional Architecture:** Built using robust, industry-standard technologies (Spring Boot, MySQL, Spring Security) ensuring high scalability and data safety.

LIMITATIONS:

Offline Access: The application currently requires an active server connection; offline mode with local data synchronization is not yet implemented.

Limited Document Editing: While optimized for viewing PDFs and PPTs, the system does not currently support real-time editing of the original uploaded files within the browser.

Platform Restriction: The system is currently designed as a responsive web application; a dedicated native mobile app (Android/ iOS) is a planned future expansion.

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Thank you

